CSE 3411

System Analysis and Design (SAD)



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SDLC Steps

i. Idea Generation (optional)
ii. Planning
iii. System Information/Requirement Gathering
iv. Analysis
v. Design
vi. Development/Construction/Coding
vii. Testing
viii. Deployment
ix. Post–Implementation & Maintenance

Requirements Analysis

- Information Sources
- Information Gathering
- Comparison
- Gap Analysis
- Feature List Fixation

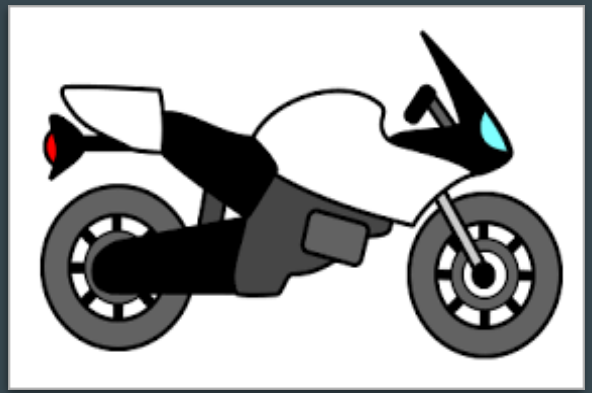
What is Requirements Analysis?

1. Process of gathering and defining the requirements for a software system
2. Goal is to ensure that the software development process is based on a clear, concise, correct and comprehensive understanding of the customer's needs and requirements
3. Involves stakeholders from various levels relevant to the project

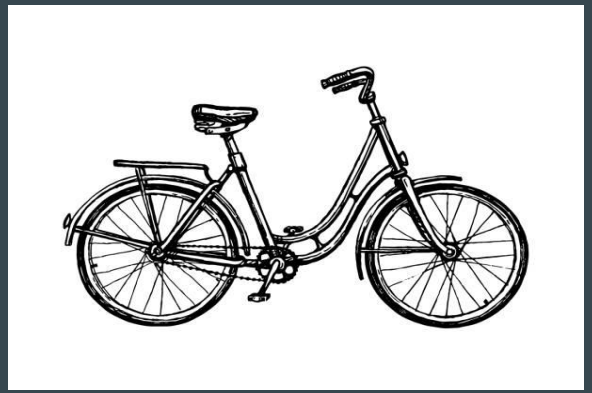
Why do we need this phase?

Sometimes (a lot of times actually), the end-product does not fully comply with what the customer/end-user had expected.

Client Expectation



Client Receives



Why do we need this phase?

- **Compliance with Business Objectives**
- **User and Product Expectations Identified**
- **Compliance and Regulation Requirements**

Types of Information

1. About Organization
2. About Human Resources
3. About Works

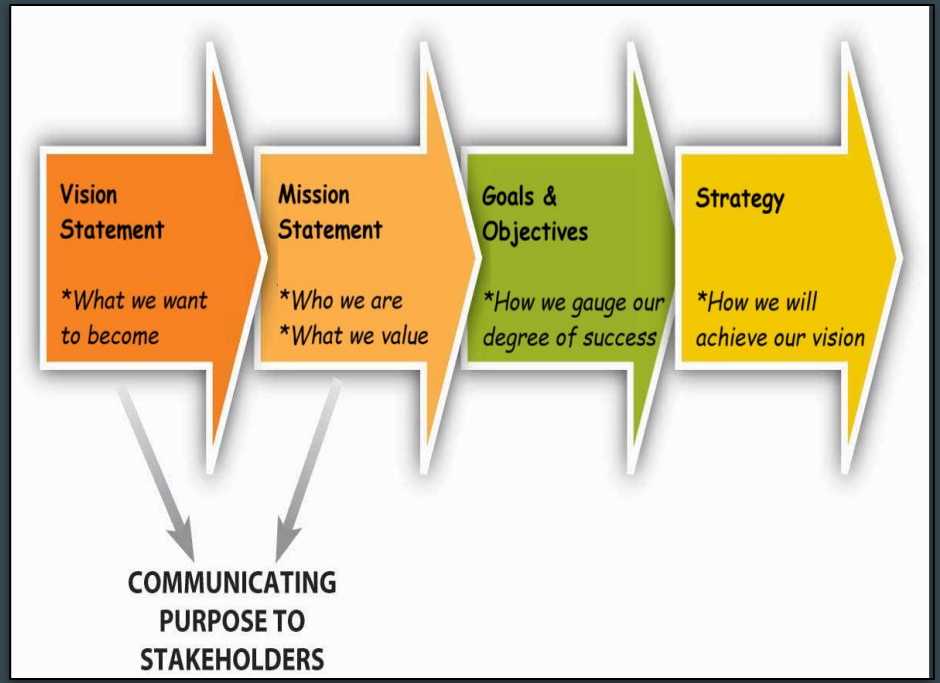
Types of Information

1. About Organization

- **Vision** - states what the organization aspires to become in the future
- **Mission** - reflects the organization's past and present by stating why the organization exists and what role it plays in society
- **Goals** - more specific aims that organizations pursue to reach their visions and missions
- **Policies** - code of conduct, confidentiality agreement, morality clause, etc
- **Organizational Structure** - roles and responsibilities

Types of Information

1. About Organization





Types of Information

2. About Users

- Type of user
  (Manager/Executive/Supervisor etc)
- Job Responsibilities
  (Role Management)
- Interpersonal Relationships

Types of Information

3. About Works

- List of works/activities
- Process flow of works/tasks
- Methods/Procedures/Business rules
- Work Schedule

Roles & Job Responsibilities

Who are the most common types of users, and what roles do they have in the company?



Responsibles for Information Gathering

Business Analysts, System Analysts, Product Owners and Project Managers are mainly responsible for meeting with stakeholders and clients, and jotting down the project's requirements, goals and scope.

Major Steps of **Information Gathering**

1. Identifying the sources of information
2. Determining the effective tools
3. Using best practices of the different tools

Sources of Information (major types)

1. Internal Sources (for existing organization)
   - Existing/official documentations
   - Personnel / Staff
   - Existing users / customers

2. External Sources
   - Benchmark products
   - Online reports / write-ups
   - Research papers / Online articles
   - Related authorities / experts

Major Sources of Information

- Benchmark Products Study
- Newspapers & Professional Journals
- Personal Staff
- Professional/ Consultant
- System Documentation
- The Users
- Reports and Transaction Document
- Vendors/ Suppliers
- Government Documents
- Online

Categorize the following sources of information

- Benchmark Products Study - External
- Newspapers & Professional Journals - External
- Personal Staff - Internal
- Professional/ Consultant – Internal/External
- System Documentation- Internal
- The Users – Internal/External
- Reports and Transaction Document - Internal
- Vendors/ Suppliers - External
- Government Documents - External
- Online - External

Information Gathering Tools

- Review Documents, Procedures, Forms, etc.
- On-site Observation
- Interview
- Questionnaires
- JAD (Joint Application Development)
- Online (benchmark products, literature, etc.)

Review Documents, Procedures, Forms

- First step of Information Gathering (for existing organizations)
- Going through Company Documentations, Manuals, Rules, Forms, Meeting Minutes, Government Rules, Yearly Reports etc.
- Will be able to find the General Standards of the system.

Review Documents, Procedures, Forms

Merits

- Very useful for initial concept
- Up-to-date manual saves hours of time.
- It can determine the basic requirements
- Printed forms are useful in UI designing

Demerits

- Outdated Manuals can not represent the current system
- Can not deliver all the required information on a system
- Time-consuming
- Need clearance / Need to maintain confidentiality

On-Site Observation



On-Site Observation

- Recognizing and noting people, objects and the environment along with the activities
- Major objective is to get as close as possible to the real system being studied
- Analyst (the one tasked with the on-site observation) is expected to be detached from the system which is being observed.
- Carried out in-person, through video recordings, labs, etc.
- Obtrusive (when respondents know that they are being observed)
  VS
  Unobtrusive (respondents do not know that they are being observed)

On-Site Observation

Merits

- Real-time Information
- Allows a closer connection with the organization
- Exact problem could be identified

Demerits

- Attitudes and motivations may not be observed properly, rather, only the actions are observed
- In-depth analysis is missing
- Respondents may behave differently due to presence of observer
- Time-consuming

Interviews

- A face-to-face situation in which a person (interviewer) asks a designated person being interviewed (interviewee) relevant questions which are designed to gather information about a problem area



Interviews

Merits

- Interviewing the right person for analyzing the system is a very fast process.
- An effective technique for eliciting information about the complex subjects.
- Its flexibility makes the interview a superior technique for exploring areas where not much is known.
- Can have follow-up questions based on previous responses

Demerits

- Long preparation time
- Takes long time to conduct
- Need skilful person to conduct
- Several interviews each with several people, involving more money and time

Interviews

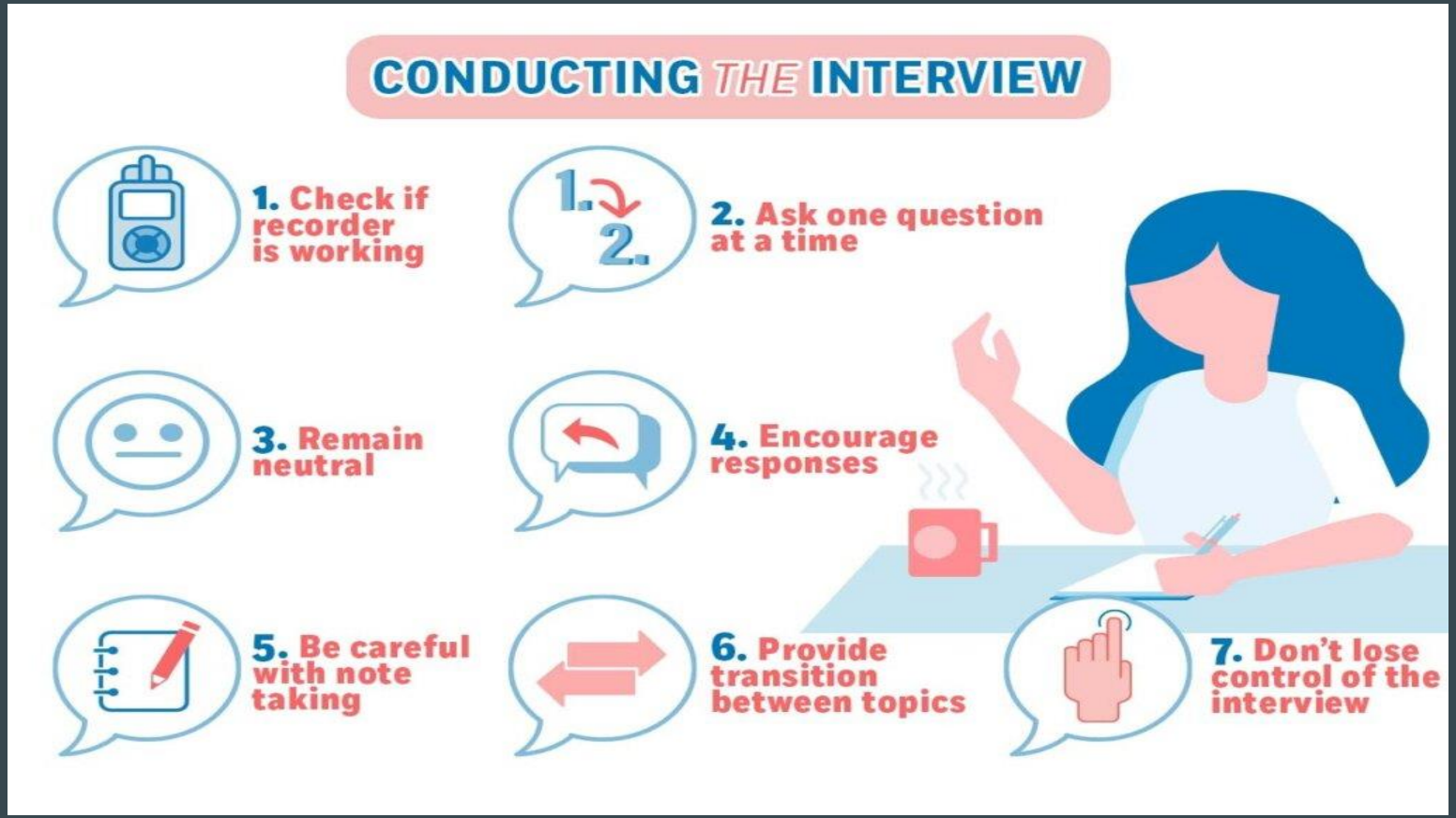
Steps To Conduct A Good Interview:

1. Select Interviewees
   - ➢Can be any important stakeholders
   - ➢Learn about the interviewee's background prior to the interview
1. Prepare for the interview
   - ➢Familiarize yourself with the agenda
   - ➢Prepare a checklist of questions
1. Conduct the interview
   - ➢Summarize the problem
   - ➢Provide incentive for the interviewee to respond
   - ➢Collect the responses
1. Summarize the interview session

Interviews


PREPERATION
for
INTERVIEW
1. Have no distractions
2. Explain interview purpose
3. Address terms of confidentiality
4. Explain interview format
5. Indicate interview duration
6. Explain how to get in touch
7. Ask if they have questions
8. Ask for permission to record

Interviews


CONDUCTING THE INTERVIEW
1. Check if recorder is working
1.
2.
2. Ask one question at a time
3. Remain neutral
4. Encourage responses
5. Be careful with note taking
6. Provide transition between topics
7. Don't lose control of the interview

Common Tips To Conduct Interviews

- Don’t jump into asking questions. Greet the interviewee and introduce yourself. Set the stage gradually.
- Establish rapport (Be welcoming). Put the interviewee at ease.
- Know what to ask and what not to.
- Phrase the questions clearly and concisely.
- Listen more, talk less.
- Take notes, audio/video recording, if possible and if allowed.
- Ask open-ended questions (don’t give choices, rather, expect long-form answers)
- Don’t ask leading/loaded questions (questions structured in a way that tend to shift the opinion of the interviewee into a certain direction)
- Avoid threatening questions
- Converge the interview with a summary with reviewing feedbacks.

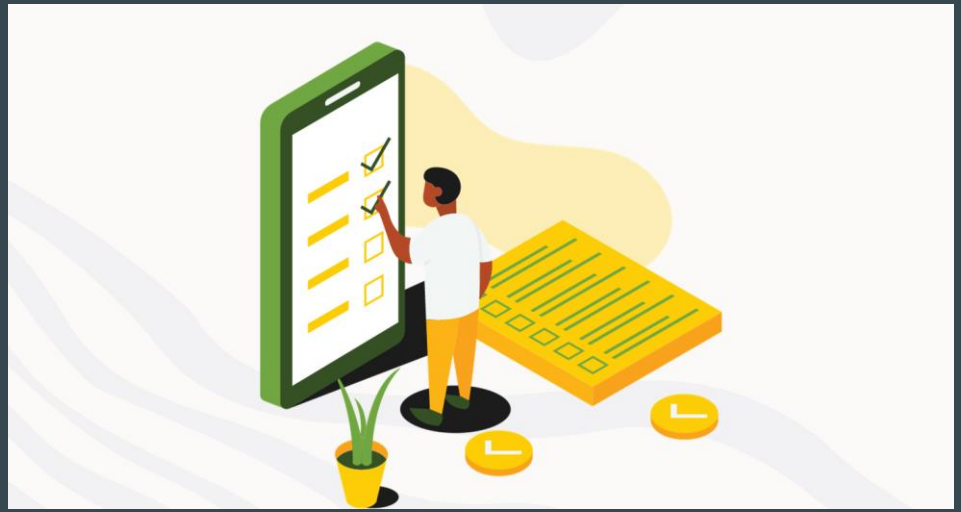
Sample Guide To Conduct Interviews

Interviewee:	Jeff Bentley, Accounts Receivable Manager
Date:	January 19, 2003
Time:	1:30 P.M.
Place:	Room 223, Admin. Bldg.
Subject:	Current Credit-Checking Policy

Time Allocated	Interviewer Question or Objective	Interviewee Response
1 to 2 min.	**Objective** Open the interview: • Introduce ourselves • Thank Mr. Bentley for his valuable time. • State the purpose of the interview — to obtain an understanding of the existing credit-checking policies.	
5 min.	**Question 1** What conditions determine whether a customer's order is approved for credit? **Follow-up**	
5 min.	**Question 2** What are the possible decisions or actions that might be taken once these conditions have been evaluated? **Follow-up**	
3 min.	**Question 3** How are customers notified when credit is not approved for their order? **Follow-up**	
1 min.	**Question 4** After a new order is approved for credit and placed in the file containing orders that can be filled, a customer might request that a modification be made to the order. Would the order have to go through credit approval again if the new total order cost exceeds the original cost? **Follow-up**	

Questionnaires

- A list of questions or items used to gather data from respondents about their attitudes, experiences, or opinions
- Can collect both qualitative and quantitative data



Questionnaires

Merits

- Economical and requires less skill to administer
- Maintains anonymity
- Can target mass audience
- Easier data collection and analysis

Demerits

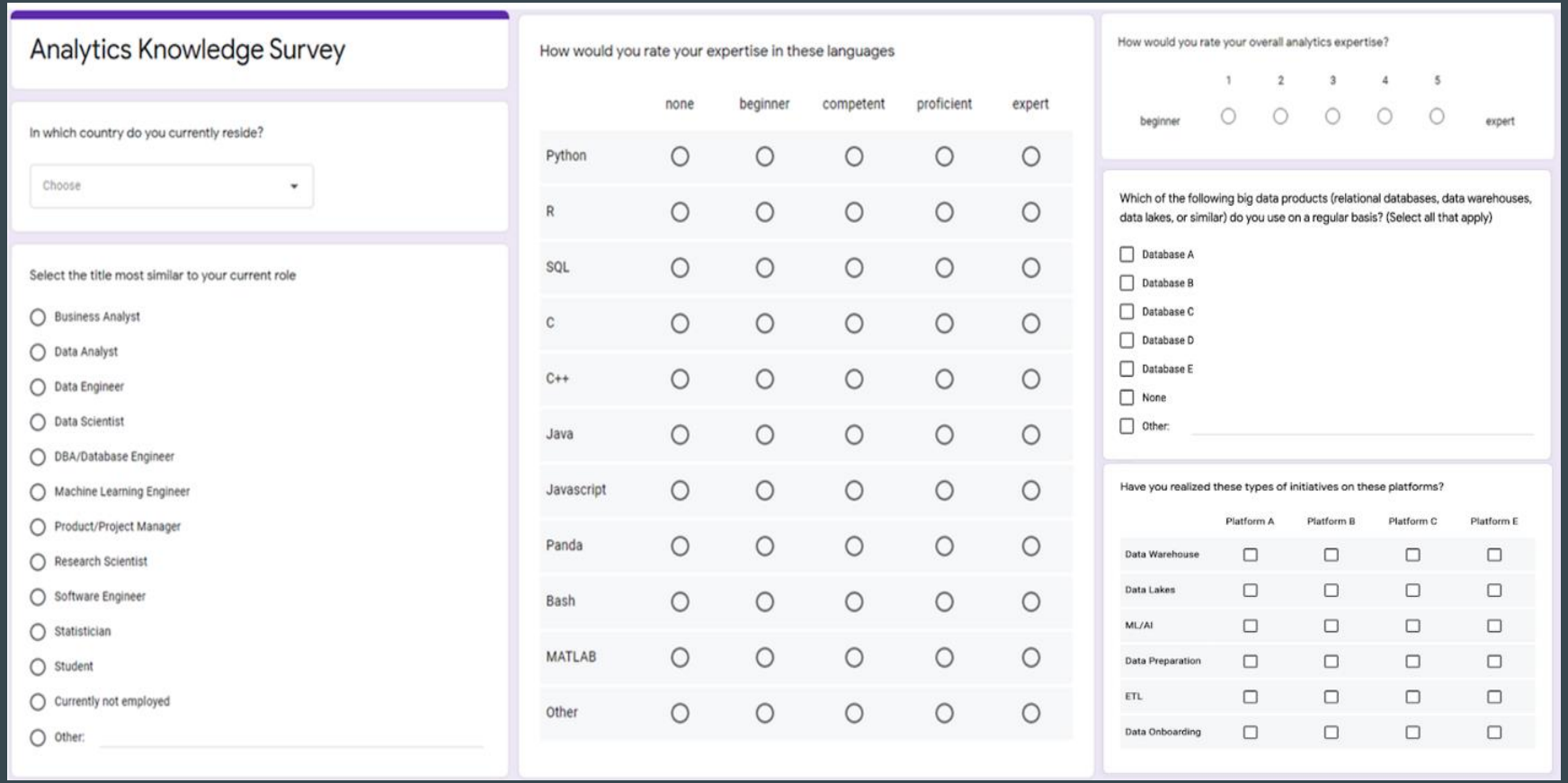
- Low response rates
- Includes bias and subjectivity
- Mostly short answers (No elaboration)
- No option to clarify doubts on both ends
- Requires knowledge of possible answers ('Other' provides no significant information)
- Uninterested participants or participants with insufficient knowledge will participate anyway, resulting in inaccurate data

Questionnaires

Types of questions

- Short Answers (one-line or one-word answers)
- Long Answers (paragraph) [not suggested]
- Multiple Choice Questions
- True/False
- Fill in the blanks
- Checkboxes
- Linear Scale (1 to 5)

Sample Requirement Analysis Survey Questions



Analytics Knowledge Survey

In which country do you currently reside?

Choose

Select the title most similar to your current role

- Business Analyst
- Data Analyst
- Data Engineer
- Data Scientist
- DBA/Database Engineer
- Machine Learning Engineer
- Product/Project Manager
- Research Scientist
- Software Engineer
- Statistician
- Student
- Currently not employed
- Other:

How would you rate your expertise in these languages

	none	beginner	competent	proficient	expert
Python					
R					
SQL					
C					
C++					
Java					
Javascript					
Panda					
Bash					
MATLAB					
Other					

How would you rate your overall analytics expertise?

	1	2	3	4	5	
beginner						expert

Which of the following big data products (relational databases, data warehouses, data lakes, or similar) do you use on a regular basis? (Select all that apply)

- Database A
- Database B
- Database C
- Database D
- Database E
- None
- Other:

Have you realized these types of initiatives on these platforms?

	Platform A	Platform B	Platform C	Platform E
Data Warehouse				
Data Lakes				
ML/AI				
Data Preparation				
ETL				
Data Onboarding				

Joint Application Development

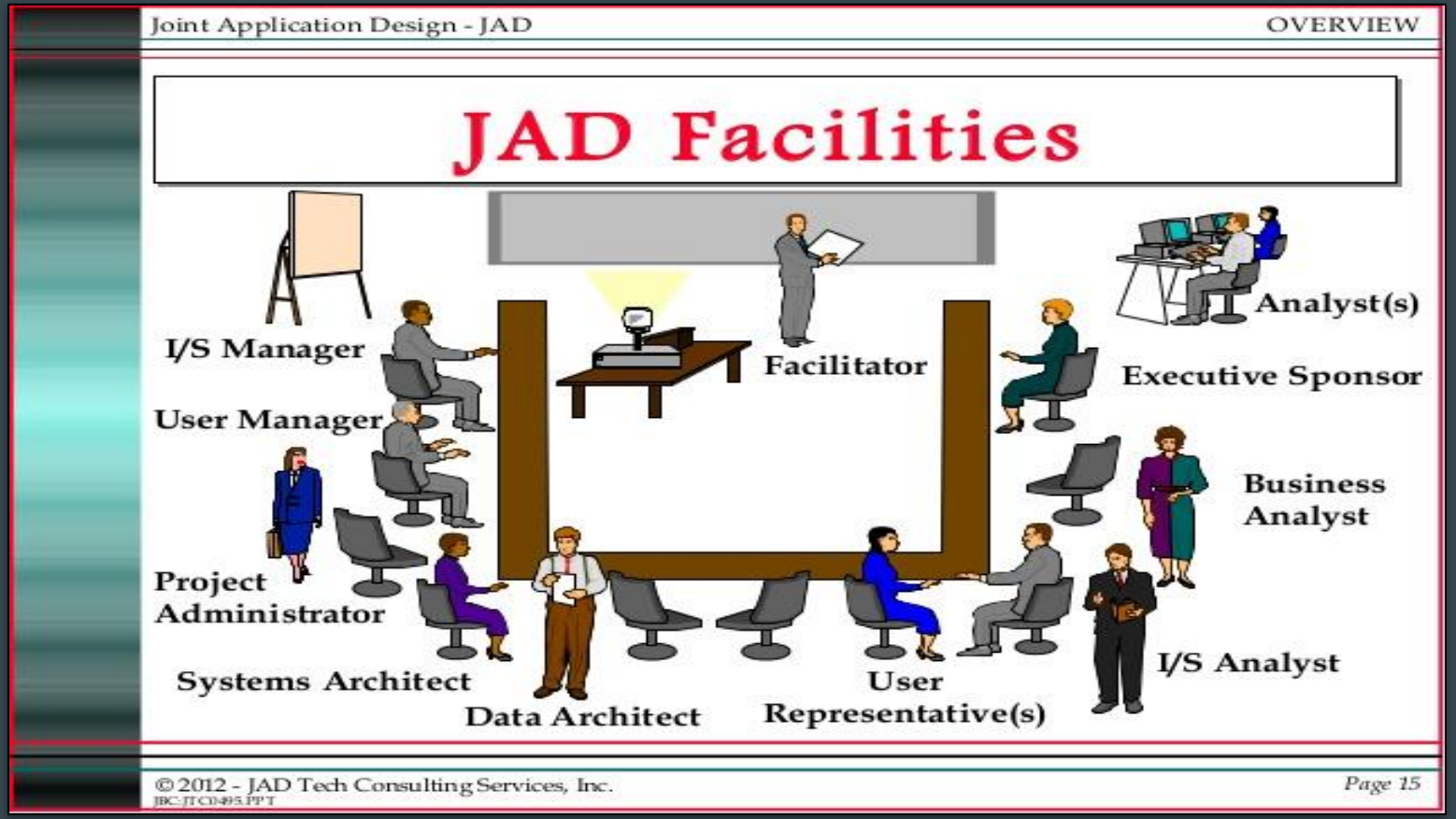
- A methodology that involves the client or end user in the design and development of a software application through a succession of collaborative workshops called JAD sessions. Its aim is to involve the customer (or end user) in defining the business needs and developing a solution based on those needs.
- It emphasizes a team-oriented development approach along with a group consensus-based problem-solving model
- Used when group consensus matters more than individual input

Joint Application Development

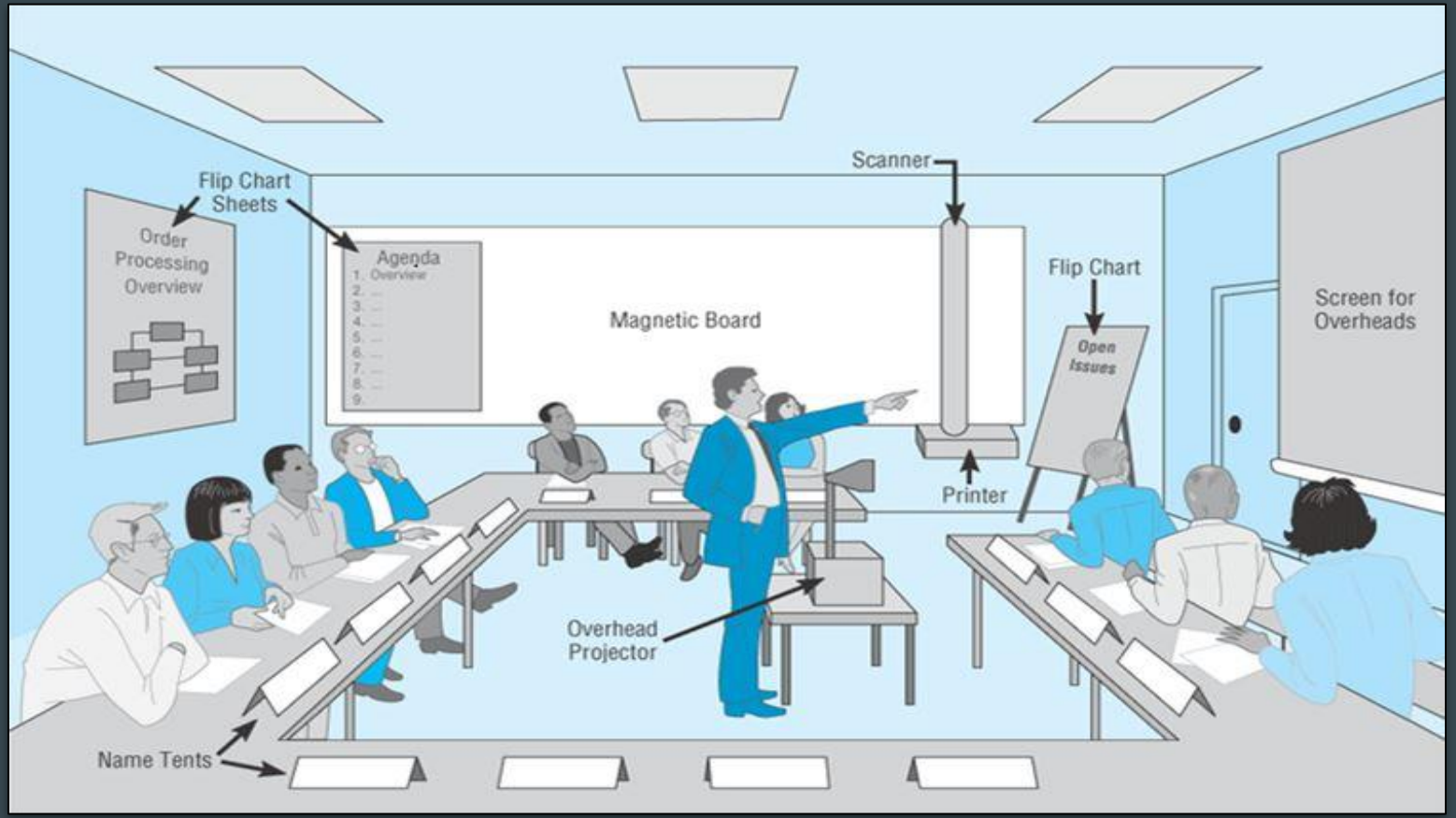
Participants of JAD sessions:

- Executive Sponsor (Someone in a decision-making capacity) such as project manager, CEO, etc
- JAD Facilitator - Manage the JAD sessions, settle disputes, encourage participation
- Users (many type of users)
- IT Representatives - Provide visualizations and prototypes
- Scribes - Document JAD sessions
- Observers - Gather knowledge (provide no input, just observe and report)

Joint Application Development



Joint Application Development

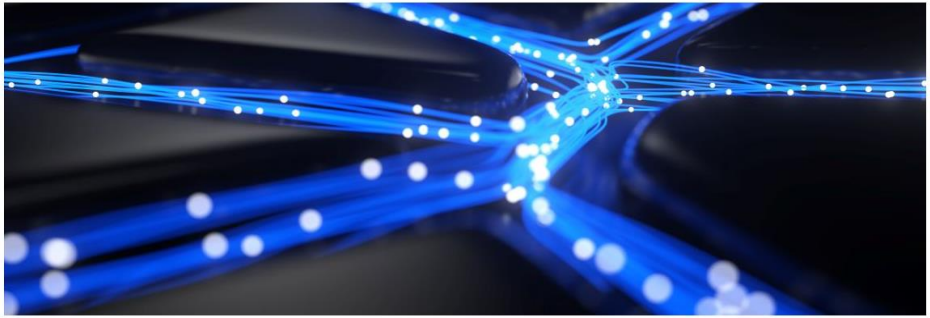


Online

- The industry standard currently in use can be identified by surfing online.
- Online articles, books, reviews, academic papers and benchmark products can help set the standard.



5 Cutting-Edge Innovations Redefining Battery Technology Right Now





RESEARCH ARTICLE Open Access

Applications of AI in classical software engineering

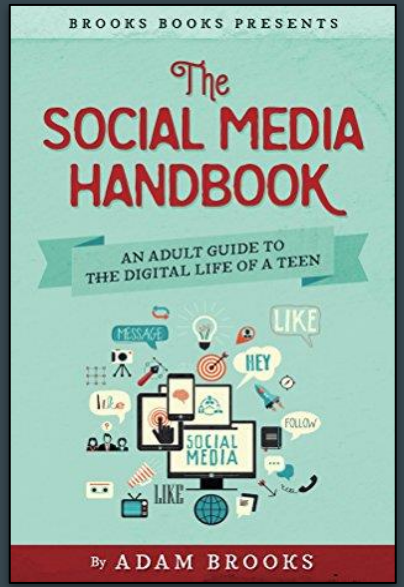
Check for updates

Marco Barenkamp[1*], Jonas Rebstadt[2] and Oliver Thomas[2]

Abstract

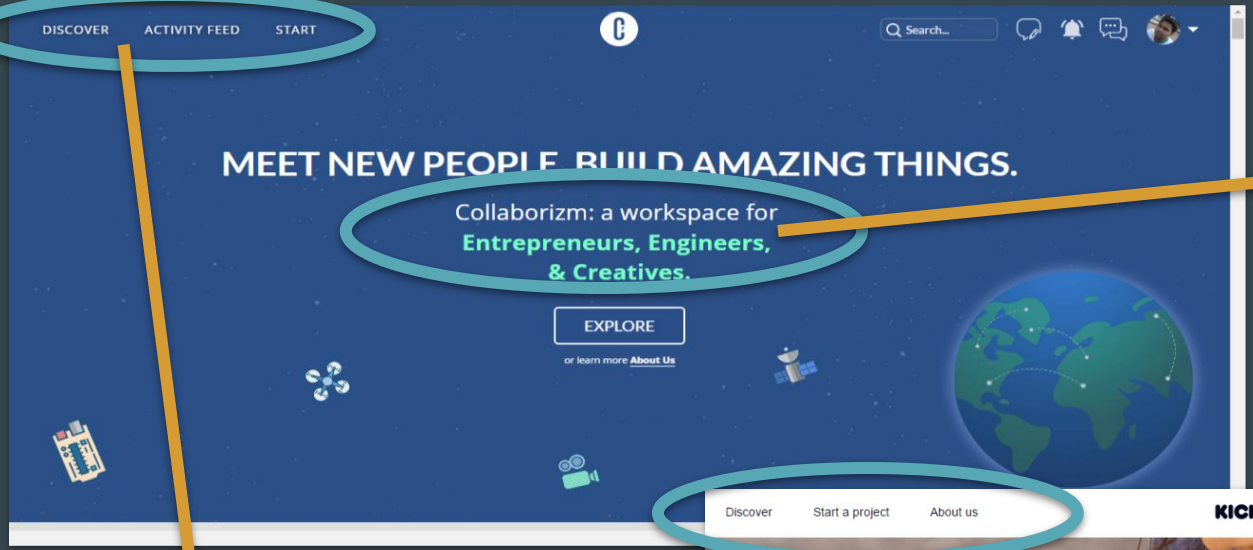
Although Artificial Intelligence (AI) has become a buzzword for self-organizing IT applications, its relevance to software engineering has hardly been analyzed systematically. This study combines a systematic review of previous research in the field and five qualitative interviews with software developers who use or want to use AI tools in their daily work routines, to assess the status of development, future development potentials and equally the risks of AI application to software engineering. The study classifies the insights in the software development life cycle. The analysis results that major achievements and future potentials of AI are a) the automation of lengthy routine jobs in software development and testing using algorithms, e.g. for debugging and documentation, b) the structured analysis of big data pools to discover patterns and novel information clusters and c) the systematic evaluation of these data in neural networks. AI thus contributes to speed up development processes, realize development cost reductions and efficiency gains. AI to date depends on man-made structures and is mainly reproductive, but the automation of software engineering routines entails a major advantage: Human developers multiply their creative potential when using AI tools effectively.

Keywords: Software engineering, AI, Artificial intelligence, ML, Machine learning, Deep learning, Pattern recognition, Neural networks



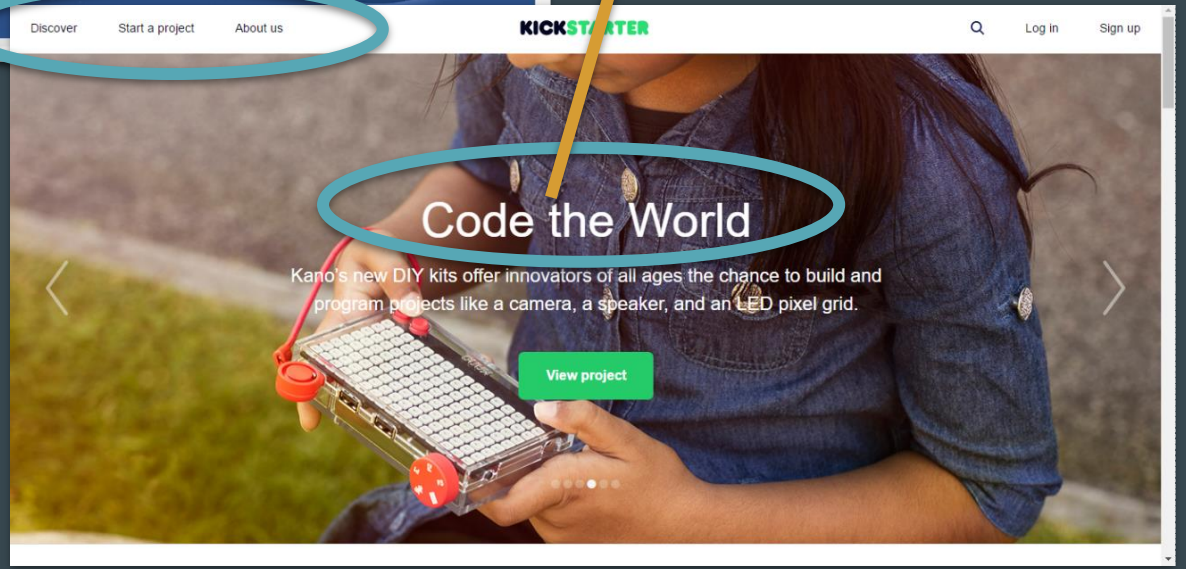
Online - Benchmark Products

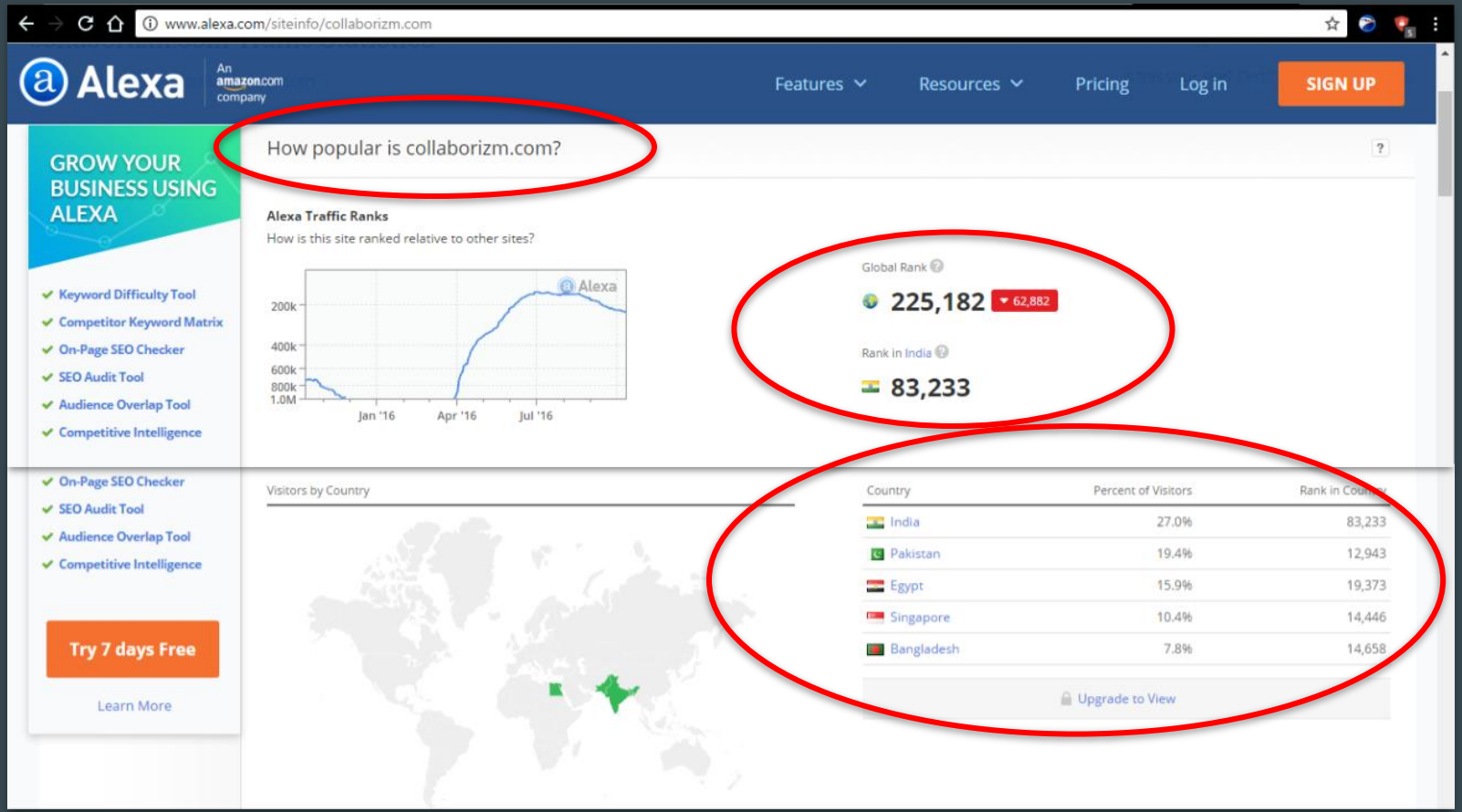
- Products of competitors
- Product benchmarking is the process of testing your products against those of your competitors, in order to find out where your product ranks in terms of consumer preference.
- Compare with past performance, identify and compare key metrics, know the industry standard



A platform for IT professionals in Bangladesh

SELECTED FEATURES




www.alexa.com/siteinfo/collaborizm.com
Alexa
An amazon.com company
Features
Resources
Pricing
Log in
SIGN UP
GROW YOUR BUSINESS USING ALEXA
Keyword Difficulty Tool
Competitor Keyword Matrix
On-Page SEO Checker
SEO Audit Tool
Audience Overlap Tool
Competitive Intelligence
How popular is collaborizm.com?
Alexa Traffic Ranks
How is this site ranked relative to other sites?
200k
400k
600k
800k
1.0M
Jan '16
Apr '16
Jul '16
Global Rank
225,182
62,882
Rank in India
83,233
Visitors by Country
Country
Percent of Visitors
Rank in Country
India 27.0% 83,233
Pakistan 19.4% 12,943
Egypt 15.9% 19,373
Singapore 10.4% 14,446
Bangladesh 7.8% 14,658
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Product Benchmarking

Merits

- Keep improving internal operations
- Understand what's working and what isn't
- Adopt or improve upon competitors' practices
- Focus on practices and offerings that promote customer satisfaction and loyalty

Demerits

- Tough to identify good and specific benchmark products
- Challenge to find reliable and pertinent data
- Copying or imitating others without taking into account your own strengths, weaknesses and uniqueness can be a risky endeavor.
- Comparing to unrealistic or unsuitable benchmarks can be disheartening and misleading

Comparison Table

- A table that uses columns for products (or services) and rows for the attributes.
- Allows for quick and easy comparison between each offering's features and characteristics.

	Product A	Product B	Product C
Factor 1	✓	✗	✗
Factor 2	—	—	✓
Factor 3	—	✓	—

Comparison Table - Steps To Create

- Select a subject to benchmark
- Decide which organizations or companies you want to benchmark
- Document your current processes
- Collect and analyze data
- Measure your performance against the data you've collected

Comparison Table - When You Don’t Need It

- Not mutually exclusive (a consumer probably doesn’t need a comparison table to help them choose a dress, because they can buy multiple dresses from multiple stores)
- Simple (coffee mug doesn’t have many meaningful, comparable attributes)
- Cheap or replaceable (not worth the time comparing the attributes of a pen)
- Unique or hard to compare

Benchmark Product









Comparison	Cricinfo	Cricbuzz	Australian Cric	IPL	Proposed system (BCL)
Team Name	✓	✓	✓	✓	✓
News	✓	✓	✓	✓	✓
Result	✓	✓	✗	✓	✓
Fixture	✓	✓	✓	✗	✓
Live Scores	✓	✓	✓	✗	✓
Login	✓	✓	✓	✗	✓
Search	✓	✓	✓	✓	✓
Blog	✓	✓	✓	✗	✓
Videos	✓	✓	✓	✓	✓
Ticket	✗	✗	✓	✗	✓
Point table	✓	✓	✓	✓	✓

Gap Analysis

- A method of assessing the performance of a business unit to determine whether business requirements or objectives are being met or having some gap with respect to expectation
- What steps should be taken to meet those gaps to positioning a product

What's in a Gap Analysis template?

- The current state
- The future state
- Gaps and their descriptions
- Next steps, proposals, and justification

Feature List Fixation

- Try to Cover All possible **Features found in Gap Analysis**
- Try to **Cover All possible Standard Features** of existing Benchmark Products

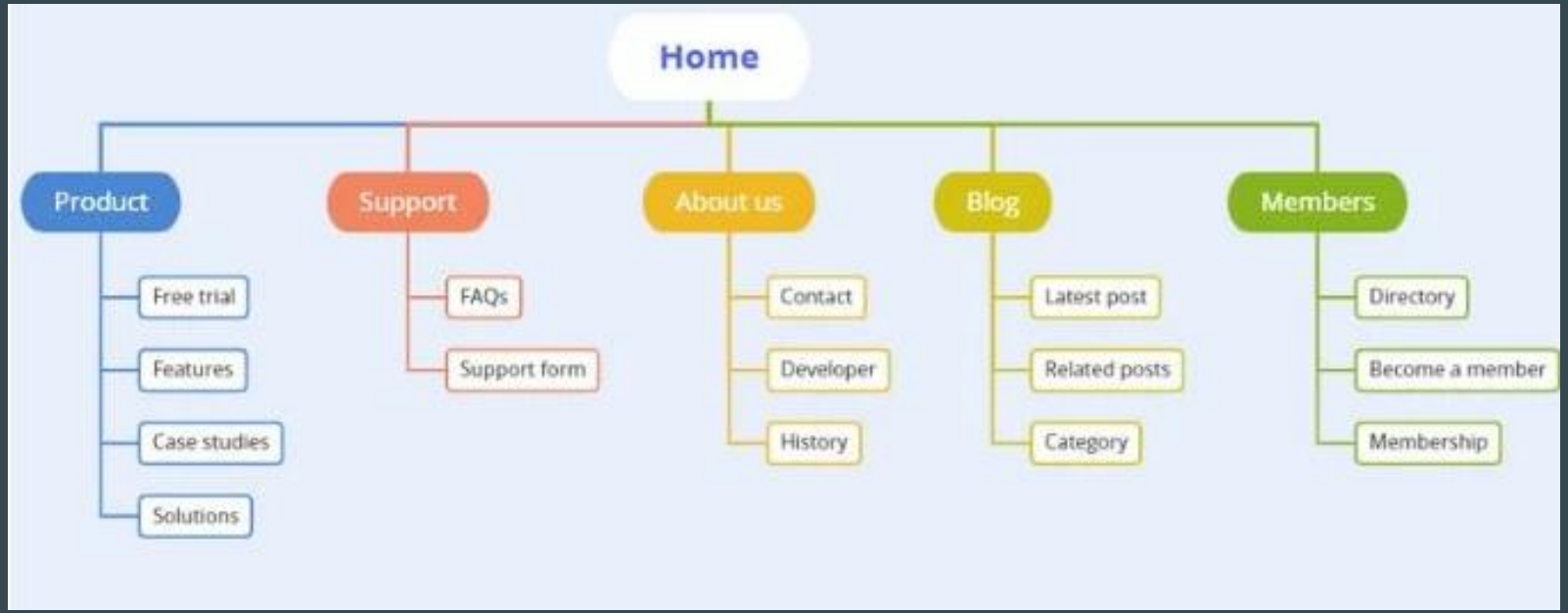
Feature List Fixation - Example

- Team Name (International , Domestic , BPL)
- Result (Result of the matches)
- Fixture (The time schedule of matches)
- Live Score (Live scores of running matches)
- Login (You can LOGIN as an Admin and an user)
- Search (You can search through Matches , Teams , Player name)
- Series (Which series are going on now)
- Point Table (The point table of Teams)
- Ticket (Online E-Ticket system for booking)
- Videos (Videos of the matches , toss , wickets , Highlights etc.)
- News (News about matches , series etc.)
- Photos (Photos of matches , Players etc.)

Feature List Fixation - With Respect To Stakeholders

- Admin
  - Activity 1
  - Activity 2
- User
  - Activity 1
  - Activity 2
- Guest
- Sponsors

Feature List Fixation - Sitemap



Feature List Fixation - Sitemap

